A nonabelian Fourier transform for unipotent representations of p-adic groups

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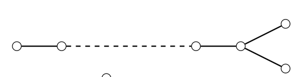
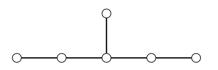
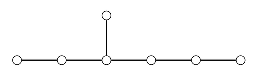
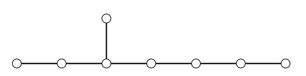
Main Aims

Find stable combinations of characters of unipotent representations of simple p-adic groups inside an L-packet.

The p-adic groups

Reductive groups over a p-adic field (commonly denoted as p-adic groups) are certain subgroups of $GL_n(F)$ with "nice" structural and representation-theoretic properties, where F is a finite extension of $\mathbb{Q}_p$. We assume that the p-adic group G is simple and split. To simplify exposition, we also assume that G is simply connected. Throughout, G refers to the simply connected p-adic group, while $G^\vee$ is its complex dual group, of adjoint type.

These groups can be classified by their root data, often visualized through their Dynkin diagrams.

Type	Dynkin diagram	G	$G^\vee$
A_n		$\mathrm{SL}_{n+1}(F)$	$\mathrm{SL}_{n+1}(\mathbb{C})$
B_n		$\mathrm{Spin}_{2n+1}(F)$	$\mathrm{PGSp}_{2n}(\mathbb{C})$
C_n		$\mathrm{Sp}_{2n}(F)$	$\mathrm{SO}_{2n+1}(\mathbb{C})$
D_n		$\mathrm{Spin}_{2n}(F)$	$\mathrm{SO}_{2n}(\mathbb{C})$
E_6		$E_6(F)_{\mathrm{sc}}$	$E_6(\mathbb{C})_{\mathrm{ad}}$
E_7		$E_7(F)_{\mathrm{sc}}$	$E_7(\mathbb{C})_{\mathrm{ad}}$
E_8		$E_8(F)$	$E_8(\mathbb{C})$
F_4		$F_4(F)$	$F_4(\mathbb{C})$
G_2		$G_2(F)$	$G_2(\mathbb{C})$

The Local Langlands Correspondence

The Local Langlands Correspondence (LLC) predicts that irreducible smooth admissible **complex** representations of a p-adic group $G = G(F)$ are controlled by geometric data associated to the dual group $G^\vee$ and the Weil group W_F of F.

More precisely, the LLC posits a bijection between:

$$\left\{\begin{matrix}\text{Irreducible smooth}\\ \text{admissible complex}\\ \text{reps of } G(F)\end{matrix}\right\} \leftrightarrow \left\{\begin{matrix}\text{Pairs } (\varphi,\phi),\ \text{where}\\ \varphi : \mathrm{SL}_2(\mathbb{C}) \times W_F \to G^\vee(\mathbb{C}),\\ \phi \in \mathrm{Irr}\left(Z(\mathrm{Im}(\varphi))/Z(\mathrm{Im}(\varphi))^0\right)\end{matrix}\right\}$$

satisfying some natural conditions (e.g., compatibility with local class field theory, preservation of L- and epsilon-factors, etc.).

Parahoric subgroups and reductive quotients

We can associate, to the p-adic group G, a polysimplical complex called the **Bruhat-Tits building** $\mathcal{B}(G)$, together with a nice action by G such that

1. For any $x \in \mathcal{B}(G)$, Bruhat and Tits construct two open compact subgroups G_x and G_x^+. The groups G_x are called **parahoric subgroups**. These are central objects in the representation theory of G. In our simplified case, $G_x = \mathrm{Stab}_G(x)$.
2. $G_x^+ \trianglelefteq G_x$ and the **reductive quotient** $G_x/G_x^+ = \mathrm{G}_x(\mathbb{F}_q)$ is a finite group of Lie type.
3. For any $g \in G$ and $x \in \mathcal{B}(G)$, $G_{g\cdot x} = gG_xg^{-1}$ and $G_{g\cdot x}^+ = gG_x^+g^{-1}$.
4. When G is simply connected, there is a bijection
$$\left\{\begin{matrix}G\text{-conjugacy classes of}\\ \text{parahoric subgroups}\end{matrix}\right\} \leftrightarrow \left\{\begin{matrix}\text{Proper subsets of the}\\ \text{affine Dynkin diagram}\end{matrix}\right\}$$

Example. If $G = \mathrm{SL}_2(\mathbb{Q}_p)$, the Bruhat-Tits building $\mathcal{B}(\mathrm{SL}_2(\mathbb{Q}_p))$ is a regular tree of degree $p+1$. Let x_0 a vertex, x_2 an adjacent vertex and x_1 the midpoint of the vertex joining x_0 and x_2.

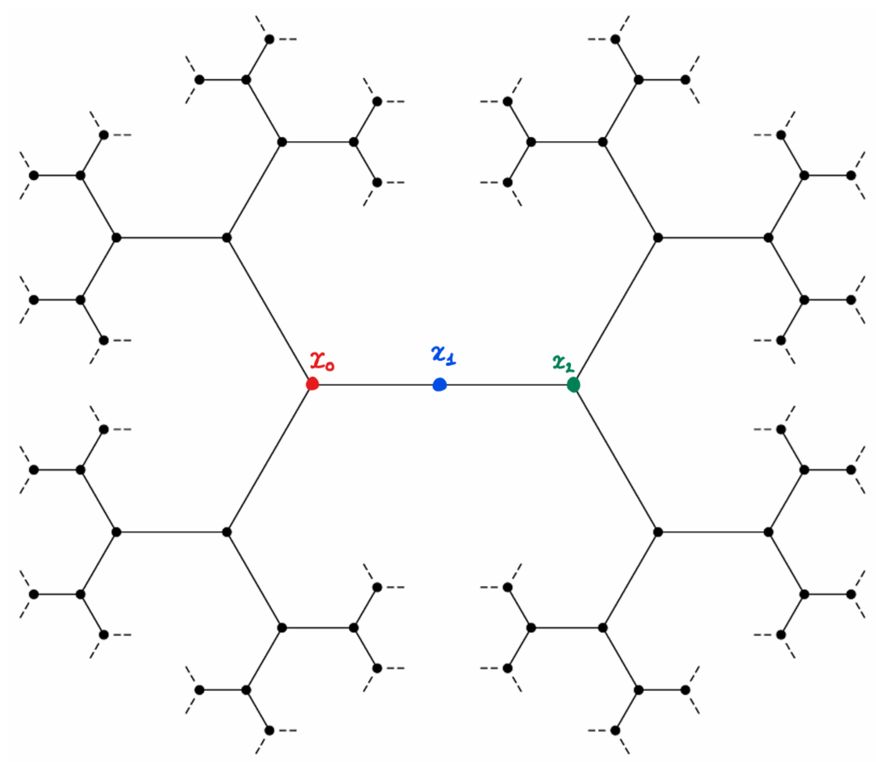


Figure 1: Bruhat-Tits building of $\mathrm{SL}_2(\mathbb{Q}_2)$

Parahoric subgroups of $\mathrm{SL}_2(\mathbb{Q}_p)$ are conjugate to one of the following open compact subgroups:

$$G_{x_0} = \begin{pmatrix}\mathbb{Z}_p & \mathbb{Z}_p\\ \mathbb{Z}_p & \mathbb{Z}_p\end{pmatrix} \qquad G_{x_0}^+ = \begin{pmatrix}1+p\mathbb{Z}_p & p\mathbb{Z}_p\\ p\mathbb{Z}_p & 1+p\mathbb{Z}_p\end{pmatrix}$$

$$G_{x_1} = \begin{pmatrix}\mathbb{Z}_p^\times & \mathbb{Z}_p\\ p\mathbb{Z}_p & \mathbb{Z}_p^\times\end{pmatrix} \qquad G_{x_1}^+ = \begin{pmatrix}1+p\mathbb{Z}_p & \mathbb{Z}_p\\ p\mathbb{Z}_p & 1+p\mathbb{Z}_p\end{pmatrix}$$

$$G_{x_2} = \begin{pmatrix}\mathbb{Z}_p & p^{-1}\mathbb{Z}_p\\ p\mathbb{Z}_p & \mathbb{Z}_p\end{pmatrix} \qquad G_{x_2}^+ = \begin{pmatrix}1+p\mathbb{Z}_p & \mathbb{Z}_p\\ p^2\mathbb{Z}_p & 1+p\mathbb{Z}_p\end{pmatrix}.$$

The corresponding reductive quotients are

$$\mathrm{G}_{x_0}(\mathbb{F}_q) \cong \mathrm{SL}_2(\mathbb{F}_q), \mathrm{G}_{x_1}(\mathbb{F}_q) \cong \mathbb{F}_q^\times, \text{ and } \mathrm{G}_{x_2}(\mathbb{F}_q) \cong \mathrm{SL}_2(\mathbb{F}_q).$$

The parahoric restriction map

For any parahoric subgroup G_x and irreducible admissible representation (π, V), the space of G_x^+-invariants $V^{G_x^+}$ is naturally a representation of the reductive quotient $\mathrm{G}_x(\mathbb{F}_q)$. This construction can be extended to a functor

$$\mathrm{res}^{G_x} : R(G) \longrightarrow R(\mathrm{G}_x(\mathbb{F}_q))$$
$$V \longmapsto V^{G_x^+},$$

where $R(G)$ and $R(\mathrm{G}_x(\mathbb{F}_q))$ are the vector spaces of formal complex linear combinations of irreducible representations of G and $\mathrm{G}_x(\mathbb{F}_q)$, respectively. This functor is called the **parahoric restriction map**.

Unipotent representations

In their groundbreaking work, Deligne and Lusztig classified the irreducible representations of finite groups of Lie type $\mathrm{G}(\mathbb{F}_q)$. They defined a special class of representations of geometric origin called **unipotent representations**, which are those that appear in the cohomology of Deligne–Lusztig varieties.

Intuition: The set of unipotent representations of $\mathrm{G}(\mathbb{F}_q)$ is a "geometric completion" of the set of principal series representations.

At the level of p-adic groups, we say that an irreducible admissible representation (π, V) of G is **unipotent** if there exists a parahoric subgroup G_x such that $\mathrm{res}^{G_x}(V)$ contains a unipotent representation of $\mathrm{G}_x(\mathbb{F}_q)$. Since principal series representations of $\mathrm{G}(\mathbb{F}_q)$ are unipotent, Iwahori-spherical representations of G are unipotent.

Unipotent representations of G have many remarkable properties. For instance, under the LLC, they correspond to pairs (φ, ϕ) where φ is trivial on the inertia subgroup I_F of W_F. Equivalently, there is a bijection between

$$\left\{\begin{matrix}\text{Unipotent irreducible}\\ \text{admissible complex}\\ \text{reps of } G(F)\end{matrix}\right\} \leftrightarrow \left\{\begin{matrix}\text{Pairs } (x,\phi),\ \text{where}\\ x \in G^\vee(\mathbb{C}) \text{ and}\\ \phi \in \mathrm{Irr}\left(Z_{G^\vee}(x)/Z_{G^\vee}(x)^0\right)\end{matrix}\right\}.$$

Moreover, the parahoric restriction functor res^{G_x} can be restricted to the space $R_{\mathrm{un}}(G)$ of virtual unipotent representations of G, giving the functor

$$\mathrm{res}_{\mathrm{un}}^{G_x} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(\mathrm{G}_x(\mathbb{F}_q))$$

whose image is the space of virtual unipotent representations of $\mathrm{G}_x(\mathbb{F}_q)$.

The nonabelian Fourier transform

Lusztig further realised that there was a basis of $R_{\mathrm{un}}(\mathrm{G}(\mathbb{F}_q))$ with geometric significance arising naturally from the theory of character sheaves on the group $\mathrm{G}(\overline{\mathbb{F}_q})$. He called these basis elements **almost characters**. The change of basis transformation between irreducible characters and almost characters of $\mathrm{G}(\mathbb{F}_q)$ is called the **nonabelian Fourier transform** denoted by

$$\mathrm{FT}^{\mathrm{G}} : R_{\mathrm{un}}(\mathrm{G}(\mathbb{F}_q)) \longrightarrow R_{\mathrm{un}}(\mathrm{G}(\mathbb{F}_q)).$$

Example. The group $\mathrm{G}(\mathbb{F}_q) = G_2(\mathbb{F}_q)$ has 10 unipotent representations, often labelled as $\phi_{1,0}$, $\phi_{1,6}$, $\phi_{2,1}$, $\phi'_{1,3}$, $G_2[1]$, $\phi_{2,2}$, $G_2[-1]$, $\phi''_{1,3}$, $G_2[\theta]$ and $G_2[\theta^2]$. The nonabelian Fourier transform fixes $\phi_{1,0}$ and $\phi_{1,6}$, while it acts on the remaining 8 representations as follows:

$$\begin{bmatrix}
\frac16 & \frac13 & \frac16 & \frac12 & \frac12 & \frac13 & \frac13 & \frac13\\
\frac13 & \frac23 & \frac13 & 0 & 0 & -\frac13 & -\frac13 & -\frac13\\
\frac16 & \frac13 & \frac16 & -\frac12 & -\frac12 & \frac13 & \frac13 & \frac13\\
\frac12 & 0 & -\frac12 & \frac12 & -\frac12 & 0 & 0 & 0\\
\frac12 & 0 & -\frac12 & -\frac12 & \frac12 & 0 & 0 & 0\\
\frac13 & -\frac13 & \frac13 & 0 & 0 & \frac23 & -\frac13 & -\frac13\\
\frac13 & -\frac13 & \frac13 & 0 & 0 & -\frac13 & \frac23 & -\frac13\\
\frac13 & -\frac13 & \frac13 & 0 & 0 & -\frac13 & -\frac13 & \frac23
\end{bmatrix}$$

Figure 2: Nonabelian Fourier transform matrix for $G_2(\mathbb{F}_q)$

Main results

It is conjectured that there is a **dual nonabelian Fourier transform** $\mathrm{FT}^\vee$ on $R_{\mathrm{un}}(G)$ for unipotent representations of G compatible with the nonabelian Fourier transform $\mathrm{FT}^{\mathrm{G}_x}$ and unipotent parahoric restriction $\mathrm{res}_{\mathrm{un}}^{G_x}$ for all $x \in \mathcal{B}(G)$. Such a map has only been constructed for a certain subspace $R_{\mathrm{un,ell}}(G)$ of $R_{\mathrm{un}}(G)$, called the space of **elliptic unipotent representations**.

Conjecture. Up to certain roots of unity, the following diagram commutes for all $x \in \mathcal{B}(G)$:

$$\begin{tikzcd}
R_{\mathrm{un,ell}}(G) \arrow[r, "\mathrm{FT}^\vee_{\mathrm{ell}}"] \arrow[d, "\mathrm{res}_{\mathrm{un}}^{G_x}"] & R_{\mathrm{un,ell}}(G) \arrow[d, "\mathrm{res}_{\mathrm{un}}^{G_x}"] \\
R_{\mathrm{un}}(\mathrm{G}_x(\mathbb{F}_q)) \arrow[r, "\mathrm{FT}^{\mathrm{G}_x}"] & R_{\mathrm{un}}(\mathrm{G}_x(\mathbb{F}_q)).
\end{tikzcd}$$

Known cases: The conjecture above holds for G of type A_n (Aubert–Ciubotaru–Romano), B_n (Moeglin–Waldspurger), G_2 (Ciubotaru–Opdam) and F_4 (L.B.). The other types remain an open problem.